
FM-AVSR Audio Reconstruction from Visual Speech: Final System Report

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fm-avsr-cleanup

TL;DR

We report the final implemented audio generation system for visual speech inputs. The delivered system converts talking-face video into speech by combining an Auto-AVSR visual encoder, SmolLM2 text hidden states, speaker identity features, and an explicit timbre/audio prompt derived from Mimi audio latents. Although the original exploration included flow-matching sampling, the best working route is a deterministic reconstruction model: a frozen base reconstructor predicts normalized Mimi latents and a residual head corrects the base prediction. The final checkpoint reaches a full validation latent correlation of 0.5818 on LRS3 with ground-truth text timestamps, and replacing ground-truth text with Auto-AVSR text only reduces correlation by about 0.0035. We also provide a raw-video inference pipeline, a Gradio interface, and a silent video mode that accepts optional reference audio for timbre control while preserving the full input video duration.

1 Introduction

The goal of this project is to synthesize intelligible speech from visual speech evidence. In the final implemented system, the input can be either a talking video with audio or a silent video with optional reference audio. The output is a waveform decoded by Mimi and, for demonstrations, an MP4 with the generated audio muxed back onto the face video.

The project began with a broader streaming lip-to-speech objective. The final deliverable is narrower and more empirical: a robust offline AVSR-conditioned audio reconstruction pipeline with explicit timbre control. This report therefore focuses only on implemented components, measured behavior, and the current engineering interface.

The final system makes four practical contributions:

1. **AVSR-compatible visual preprocessing.** We replaced the earlier RGB lip input with the correct Auto-AVSR-style grayscale mouth crop `lip_avsr.npy`, improving raw-video behavior on external examples.
2. **Deterministic residual audio-latent reconstruction.** We found that the strongest route is not stochastic flow sampling but endpoint regression in normalized Mimi latent space with a frozen base and residual correction.
3. **Explicit timbre/audio prompt conditioning.** The model uses the first 38 Mimi latent frames, about 3.04 seconds, both as prompt tokens and as global mean/std timbre statistics. A new silent-video mode can instead use a separate reference audio file for these conditions.
4. **One-command demo pipeline.** A single script now performs video standardization, face/lip preprocessing, AVSR feature extraction, SmolLM2 hidden extraction, timbre conditioning, FM-AVSR reconstruction, Mimi decode, and MP4 muxing.

2 Goal of Achievement

The achieved goal is a working visual-speech-to-audio reconstruction system with documented model checkpoints, quantitative evaluation, raw-video examples, and a usable command-line/GUI interface. The system is not claimed to be real-time and is not a production voice-cloning model; instead, it is a validated research prototype for generating Mimi audio latents from visual, text, speaker, and timbre conditions.

The final acceptance criteria were:

- generate audible speech from LRS3 validation clips and external raw videos;
- remove the first 3.04 seconds from listening examples when the same clip is used as timbre/audio prompt;
- support Auto-AVSR text as a replacement for ground-truth transcript at inference time;
- support a silent-video mode with optional reference audio while keeping the full video length;
- keep the codebase command surface focused on the active FM-AVSR route.

3 Methods

3.1 System Overview

Figure 1 summarizes the implemented pipeline. The video path first produces aligned face crops and an AVSR-compatible mouth crop. The mouth crop is encoded by Auto-AVSR into visual features and a CTC-decoded text string. The text is passed through SmoLLM2 to obtain hidden states. Speaker identity is extracted from aligned face crops. Audio conditioning is derived from either the input audio prefix or an external reference audio file.

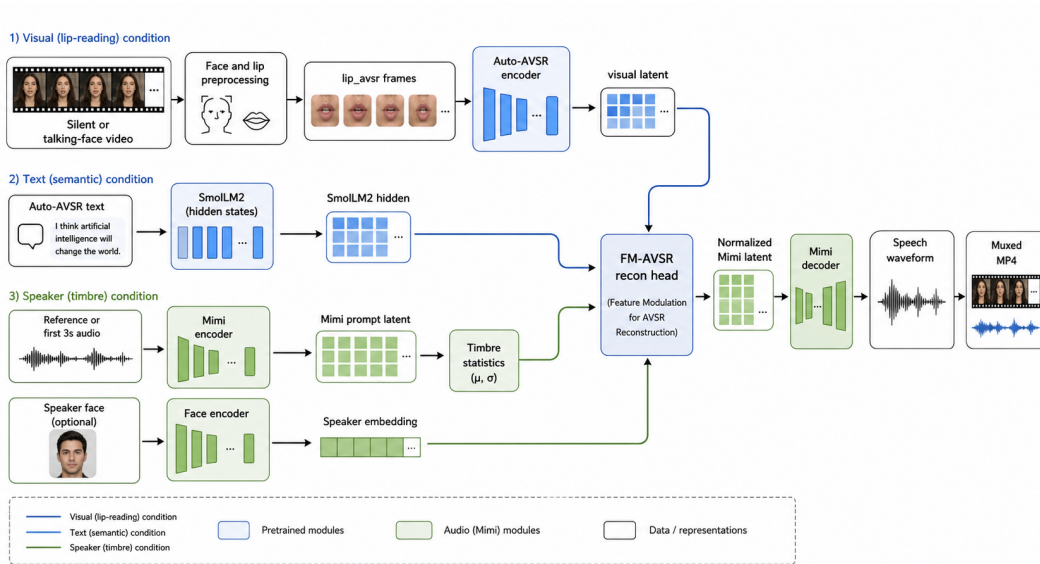


Figure 1: Implemented FM-AVSR audio reconstruction pipeline. Visual, text, speaker, timbre, and audio-prompt conditions are fused by the reconstruction model to predict normalized Mimi latents, which are decoded to waveform and muxed back to video.

3.2 Data and Latent Representation

The final route uses the following per-clip files:

File	Shape	Role
lip_avsr.npy	$(T_v, 96, 96)$	Auto-AVSR-compatible grayscale mouth crop
avsr_enc_lipavsr.npy	$(T_v, 768)$	visual condition
avsr_text_lipavsr.txt	text	Auto-AVSR transcript for raw video
smollm2_h_lipavsr.npy	$(L, 960)$	SmolLM2 hidden states for Auto-AVSR text
speaker_emb.npy	(256)	face-based speaker/identity condition
latent.npz	$(T_a, 512)$	Mimi continuous latent target or dummy target
audio_prompt.npy	$(38, 512)$	normalized Mimi prompt tokens
timbre_cond.npy	(1024)	prompt mean and standard deviation

Table 1: Core files consumed by the final reconstruction path.

Mimi latents are normalized per dimension:

$$y_t = \frac{z_t - \mu}{\sigma}, \quad z_t \in \mathbb{R}^{512}, \quad (1)$$

where z_t is the continuous Mimi latent before decoding. Metrics are computed in normalized latent space. Generated latents are denormalized and decoded by the Mimi decoder:

$$\hat{z}_t = \hat{y}_t \sigma + \mu, \quad \hat{\mathbf{x}}_{\text{wav}} = \text{MimiDecoder}(\hat{z}_{1:T}). \quad (2)$$

3.3 Condition Construction

Auto-AVSR features are produced at approximately 25 Hz, while Mimi latents are used at 12.5 Hz. We align by taking every second visual frame:

$$v_t = \text{AVSR}(V)_{2t}, \quad v_t \in \mathbb{R}^{768}. \quad (3)$$

Text hidden states are obtained by encoding either ground-truth transcript (`text_json`) or Auto-AVSR transcript (`lipavsr`) with SmolLM2. For in-distribution LRS3 evaluation, word timestamps map audio-latent frames to language-model positions. For external raw videos, no ground-truth timestamps exist, so hidden states are uniformly resampled.

The timbre condition is computed from normalized prompt latents $A \in \mathbb{R}^{38 \times 512}$:

$$q = [\text{mean}(A), \text{std}(A)] \in \mathbb{R}^{1024}. \quad (4)$$

The prompt tokens A are also provided directly as cross-attention condition tokens. In silent-video mode, A and q are derived from optional reference audio; if no reference is provided, both are zero.

3.4 Reconstruction Model

The strongest model is a deterministic endpoint reconstructor rather than a sampling model. Figure 2 contrasts the original flow-matching sampling route with the implemented route. The current path calls `reconstruct_from_cond()` with

$$\tilde{x} = 0, \quad \tau = 1. \quad (5)$$

Thus the network predicts an endpoint directly from condition C :

$$y_{\text{base}} = f_{\text{base}}(C), \quad \Delta = f_{\text{residual}}(C), \quad \hat{y} = y_{\text{base}} + \Delta. \quad (6)$$

There is no random noise input at inference time and no iterative denoising loop. The residual base is a frozen recon checkpoint, and the final checkpoint learns the correction term.

3.5 Silent Video with Reference Audio

The latest interface supports a silent input video and optional reference audio. The video is processed without requiring an original audio track. A temporary silent audio track is added only so the legacy face/audio preprocessing worker can run. The target latent is a zero dummy latent with length determined by the video duration; it is used only to set output length and metrics bookkeeping. When a reference audio file is provided, it is converted to 24 kHz mono WAV, encoded by Mimi, normalized, and used to populate `audio_prompt.npy` and `timbre_cond.npy`. Output muxing does not crop the first 3.04 seconds, so the generated MP4 preserves the full input duration.

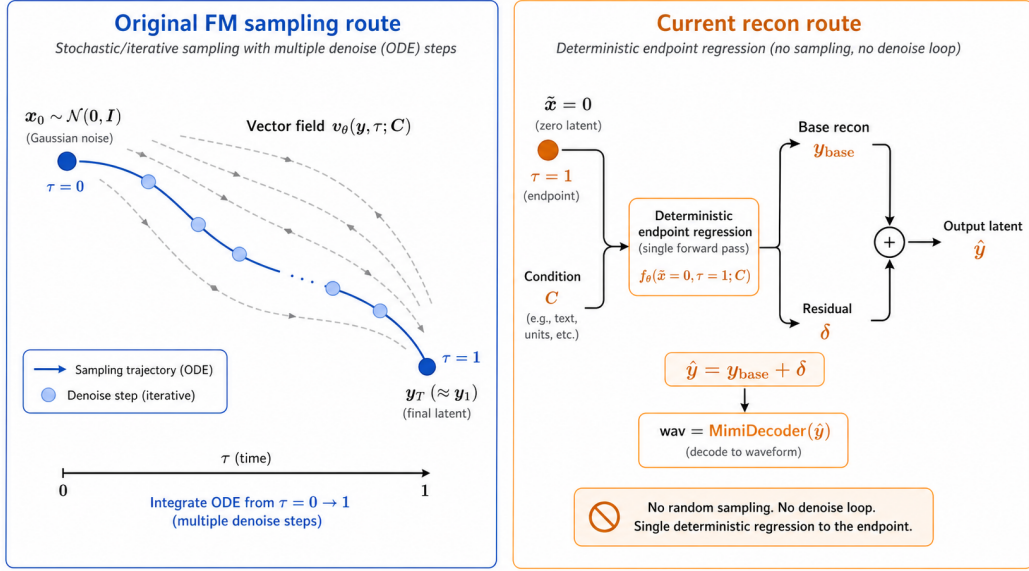


Figure 2: The final system uses deterministic residual reconstruction. The original FM sampling route learns a vector field from noise to target; the final route fixes $\tilde{x} = 0$ and $\tau = 1$, then predicts an endpoint with a frozen base plus residual correction.

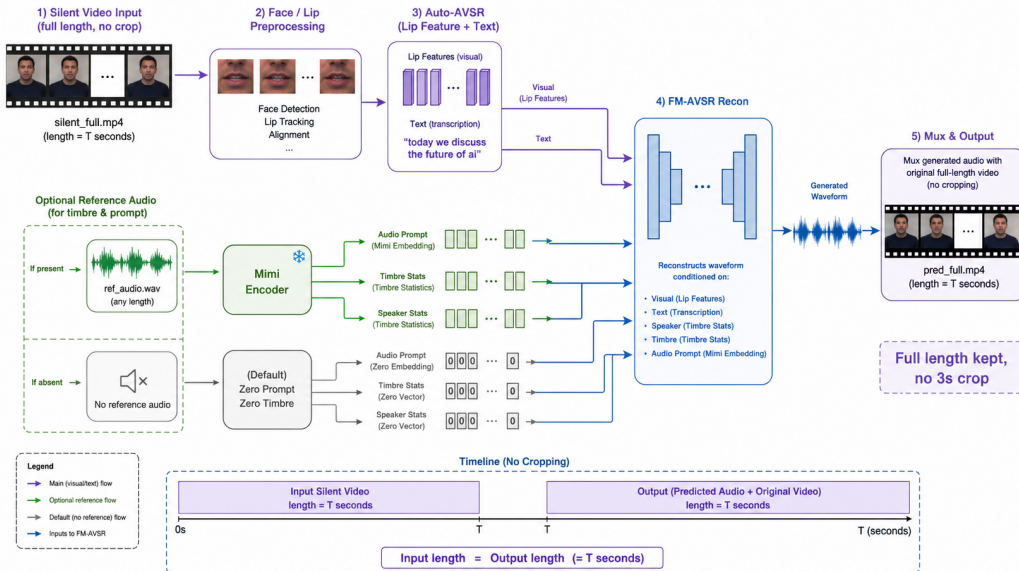


Figure 3: Silent-video inference mode. Optional reference audio controls timbre and prompt conditions. If the reference is absent, zero prompt and zero timbre are used. The output video keeps the full input duration and is saved as `<exp>_pred_full.mp4`.

4 Experiments and Results

4.1 Final Checkpoint

The final configuration and checkpoint are:

- **Config:** configs/fm_avsr_lipavsr_59144_timbre3s_audioprompt38_pool_promptstats005_residual_samplecorr02_from1000_recon_textjson_wordts.yaml
- **Checkpoint:** runs/fm_avsr/lipavsr_59144_timbre3s_audioprompt38_pool_promptstats005_residual_samplecorr02_from1000_recon_textjson_wordts_v1/step_001500.pt
- **Residual base:** runs/fm_avsr/lipavsr_59144_timbre3s_audioprompt38_pool_from4000_recon_textjson_wordts_v1/step_005000.pt

Key training choices are deterministic reconstruction loss ($\lambda_{\text{recon}} = 1.0$), sample-level correlation loss ($\lambda_{\text{corr}} = 0.2$), and prompt-timbre statistics loss ($\lambda_{\text{prompt}} = 0.05$).

4.2 Validation Metrics

Setup	Corr	MSE	MAE
Final checkpoint, text_json + word timestamps	0.58184180	0.66763106	0.60181897
Same checkpoint, Auto-AVSR text lipavsr	0.57833471	0.67162028	0.60371083

Table 2: Full LRS3 validation metrics on 1000 clips. Metrics skip the first 38 Mimi frames, corresponding to the 3.04-second prompt region.

Replacing ground-truth text with Auto-AVSR text reduces latent correlation by only 0.00351. This suggests that, in the current fusion design, the dominant factors are visual and timbre conditions rather than exact transcript quality. This conclusion should not be read as a general claim that text is useless; it only characterizes the implemented reconstruction model.

4.3 External Raw Video Checks

The raw-video path was tested on external .mov inputs. Correct Auto-AVSR-compatible cropping improved the Trump example substantially relative to the older RGB lip input:

Video	Old RGB lip path	Correct lip_avsr path
Trump	0.30353260	0.38691377
HRX	0.25592437	0.27240886

Table 3: External-video latent correlation checks. These use Auto-AVSR text and uniform text alignment because external videos do not include LRS3 word timestamps.

The Trump example also produced plausible Auto-AVSR text:

OUR COUNTRY IS WINNING AND IN FACT WE'RE WINNING SO MUCH ...

4.4 Silent-Video Reference-Audio Smoke Test

The new silent-video mode was verified with an 8-second silent Trump clip and a 3.5-second reference audio segment from the same source. The generated output kept the full duration:

Item	Value
Silent input duration	8.000 s
Generated MP4 duration	8.000 s
Dummy target latent shape	(100, 512)
Audio prompt shape	(38, 512), nonzero
Timbre condition shape	(1024), nonzero
Output	eval_out/trump_silent_ref_verify/trump_silent_ref_verify_pred_full.mp4

Table 4: Silent-video reference-audio end-to-end verification.

5 Engineering Deliverables

Command-line pipeline. The primary entry point is:

```
python scripts/run_raw_video_avsr_recon_pipeline.py \  
  --input data/trump.mov --exp trump_cleanup_verify --force
```

For silent video with reference audio:

```
python scripts/run_raw_video_avsr_recon_pipeline.py \  
  --input silent.mp4 --ref_audio ref.wav \  
  --silent_input --exp demo --force
```

GUI. `scripts/gradio_avsr_gui.py` exposes video upload, optional reference audio, and silent-input controls. It calls the same backend pipeline.

Codebase cleanup. The active `scripts/` surface now contains only the FM-AVSR route and raw-video utilities. Historical `v2/v3/v4`, `Mimi-code`, `teacher-cache`, and `sweep` scripts were moved into `archive/scripts/`.

Tests. The final branch includes focused tests for dataset text sources, evaluation helpers, timbre/audio prompt handling, and raw-video pipeline helper behavior. The final verification command ran 97 unit tests successfully.

6 Limitations

- The system is offline and not real-time. Previous timing showed that face alignment and lip cropping dominate runtime; the FM head and Mimi decoder are not the bottleneck.
- The final model is deterministic. It improves stability but can average speaker-specific acoustic detail, which partly explains the “mixed voice” artifacts observed in listening tests.
- External videos remain harder than LRS3 validation clips because of domain shift, missing word timestamps, and crop quality differences.
- The silent-video reference-audio mode controls timbre through latent prompt statistics; it is not a full speaker cloning model and does not guarantee high-fidelity identity preservation.
- Metrics in silent mode compare against a dummy zero latent and are only bookkeeping, not a quality estimate.

7 Conclusion

The final delivered system is a practical AVSR-conditioned audio reconstruction pipeline. Its main lesson is that, for the available data and model capacity, deterministic Mimi-latent reconstruction with explicit timbre/audio prompt conditioning is stronger and more reliable than the explored stochastic FM sampling route. Correct Auto-AVSR-compatible preprocessing is essential, and text-source quality has a smaller measured effect than visual and timbre conditioning in the final architecture. The resulting codebase now supports both standard talking-video demos and silent-video demos with optional reference audio, making the system suitable for final presentation and further targeted improvement.